Predicting Coronary Heart Disease with Linear Models

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Abstract

We compare ridge and regularized logistic regression for coronary heart disease classification using medical survey data. Our NumPy pipeline removes constant and highly incomplete features, imputes missing values, and standardizes the features. The best tested configuration is ridge regression with $\lambda = 10^{-6}$ and threshold 0.21, achieving validation F1 0.4134, precision 0.3266, and recall 0.5629. After retraining on all labeled data, the AIcrowd submission achieves F1 0.417 and accuracy 0.861. An always-negative baseline achieves high validation accuracy but detects no positive cases, illustrating the limitations of accuracy for this imbalanced task.

1 Introduction

We predict the supplied coronary heart disease label from health and lifestyle features originating from the Behavioral Risk Factor Surveillance System (BRFSS) [1]. The dataset contains 328,135 labeled training observations and 109,379 test observations, each with 321 features.

Only 8.83% of training observations belong to the positive class. We therefore select configurations using validation F1 and report accuracy, precision, and recall to describe the classification trade-off. We compare two linear models under a common preprocessing and validation protocol.

2 Methods

2.1 Preprocessing

In the complete training data, 239 columns contain `NaN` values, including 147 with more than 50% missing entries. Six columns are constant among their observed values; none is entirely missing.

Within the training subset, we remove columns with more than 90% missing values, entirely missing columns, and columns constant among their observed values. This retains 218 of the original 321 features on our training split, before adding the intercept.

Remaining `NaN` entries are replaced by training column medians. Features are standardized:

$$\widetilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \tag{1}$$

where means and standard deviations are computed after imputation. Zero standard deviations are replaced by one, and an intercept column is added.

Column selection and all imputation and scaling parameters are learned exclusively on training data and reused for validation. This baseline treats retained features as numerical; it does not explicitly encode categories or interpret numerical survey response codes as missing values.

2.2 Algorithms

We implement the six required methods: linear regression with gradient descent and stochastic gradient descent using batch size one, least squares, ridge regression, logistic regression, and regularized logistic regression. Each returns final weights and a loss excluding the regularization penalty.

For the predictive comparison, targets are converted to $\{0, 1\}$. Ridge minimizes

$$\frac{1}{2N}\|y - Xw\|_2^2 + \lambda\|w\|_2^2, \tag{2}$$

using `numpy.linalg.solve` to solve

$$(X^\top X + 2N\lambda I)w = X^\top y. \tag{3}$$

Logistic regression uses $p_i = \sigma(x_i^\top w)$, where $\sigma(z) = 1/(1 + \exp(-z))$, and minimizes

$$\begin{aligned}\mathcal{L}(w) = &-\frac{1}{N}\sum_{i=1}^{N}[y_i \log p_i + (1 - y_i)\log(1 - p_i)] \\ &+ \lambda\|w\|_2^2.\end{aligned} \tag{4}$$

Weights start at zero and are updated for 300 gradient descent iterations with learning rate 0.01. We use numerically stable sigmoid and loss expressions. Both regularized models penalize the intercept.

3 Experimental Setup

We use a stratified 80/20 training–validation split with `numpy.random.default_rng(42)`, shared by all configurations. For each model, we test

$$\begin{aligned}\lambda \in \{0, 10^{-6}, 10^{-5}, 5{\times}10^{-5}, 10^{-4}, \\ 5{\times}10^{-4}, 10^{-3}, 5{\times}10^{-3}, 10^{-2}\}.\end{aligned}$$

Classification thresholds range from 0.10 to 0.60 in steps of 0.01.

An observation is positive when its score is at least the threshold. Ridge uses Xw directly; logistic regression uses sigmoid probabilities. We select configurations using

$$F_1 = \frac{2TP}{2TP + FP + FN}. \tag{5}$$

Because regularization and thresholds are selected on the same validation set, the best validation score may be optimistic. AIcrowd provides a separate evaluation of the final submission.

Table 1: Selected validation results at each configuration's best tested threshold.

Model	λ	Threshold	Accuracy	F1
Always negative	–	–	0.9117	0.0000
Ridge	0	0.23	0.7694	0.2534
Ridge	10^{-6}	0.21	0.8589	**0.4134**
Ridge	10^{-5}	0.21	0.8592	0.4133
Ridge	10^{-4}	0.20	0.8495	0.4126
Ridge	10^{-2}	0.20	0.8525	0.4100
Logistic	0	0.39	0.8416	0.3988
Logistic	5×10^{-4}	0.39	0.8413	0.3991
Logistic	10^{-2}	0.40	0.8495	0.3990

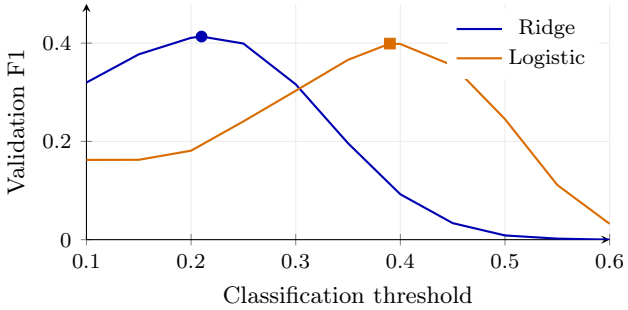


Figure 1: Validation F1 at selected thresholds for ridge with $\lambda = 10^{-6}$ and logistic regression with $\lambda = 5 \times 10^{-4}$. Markers indicate the best thresholds from the full search.

4 Results and Discussion

Table 1 summarizes selected configurations from the full grid. Complete results are saved in `validation_results.csv`.

The best tested model is ridge with $\lambda = 10^{-6}$ and threshold 0.21, achieving validation F1 0.4134, precision 0.3266, and recall 0.5629. It identifies approximately 56.3% of positive cases, while 32.7% of its positive predictions are correct.

Its validation F1 exceeds the best tested logistic configuration by 0.0143. However, logistic convergence has not been verified, so this comparison applies only to the tested optimization settings.

Positive regularization substantially improves ridge over its unregularized result. The latter should be interpreted cautiously because solving unregularized normal equations can be numerically unstable with dependent or nearly dependent features. Among positive values, F1 varies from approximately 0.4100 to 0.4134; the difference between $\lambda = 10^{-6}$ and 10^{-5} is negligible.

Threshold choice has a larger observed effect (Figure 1). For the selected ridge model, F1 increases from 0.0086 at threshold 0.50 to 0.4134 at 0.21. Ridge scores are not constrained to $[0, 1]$ and are not calibrated probabilities.

An earlier pipeline without the 90% missingness filter achieved best F1 0.4113. The final pipeline also uses a broader regularization grid and finer threshold search, so the improvement cannot be attributed solely to feature removal.

The majority baseline has higher validation accuracy but zero recall and F1. Our models sacrifice accuracy to detect positive cases. Further evaluation should isolate preprocessing effects and assess variability across repeated splits.

5 Final Model and Reproducibility

We retrain ridge with $\lambda = 10^{-6}$ and threshold 0.21 on all labeled training observations. Preprocessing is refitted on these observations, retaining 218 features, and applied to the test set. Predictions are converted back to $\{-1, 1\}$ and exported with the original identifiers using `create_csv_submission`.

The final CSV contains 109,379 predictions: 16,624 positive and 92,755 negative. The AIcrowd submission achieves F1 0.417 and accuracy 0.861, close to the validation results of 0.4134 and 0.8589, respectively.

The code provides `implementations.py` for the six methods, `preprocessing.py` for transformations, and `train.py` for experiments. The final script, `run.py`, reads the configuration with the highest validation F1 from `validation_results.csv` and generates the submission. It uses NumPy and the Python standard library.

To reproduce the final predictions, with the three dataset CSV files in `data/`, run:

```
python run.py -data data
```

Repository: `https://github.com/linakbj/ml_project1`.

6 Conclusion

Ridge achieves the highest validation F1 among the tested configurations and an AIcrowd F1 of 0.417. Threshold selection is important for minority-class detection, although false positives remain frequent. Future experiments should evaluate categorical representations, preprocessing ablations, optimization convergence, and variability across validation splits.

References

[1] Centers for Disease Control and Prevention. *BRFSS: 2015 Annual Survey Data.* `https://www.cdc.gov/brfss/annual_data/annual_2015.html`.